

QFT study #3

9.15

Contents

- 0. Recap
- 1. LSZ reduction formula
- 2. Feynman propagator
- 3. Feynman rules
 - 1. Interaction picture
 - 2. Contraction
 - 3. Position space Feynman rules
 - 4. Momentum space Feynman rules
- 4. Appendix
 - 1. LSZ reduction formula derivation
 - 2. Feynman rules of Lagrangian derivation
 - 3. Wick theorem

Recap

QFT study#1 :

Free Theory

Lagrangian

$$\mathcal{L} = \mathcal{L}_{\text{kinetic}} = -\frac{1}{2} \phi_0 (\square + m^2) \phi_0$$

EOM and free field

$$(\square + m^2) \phi_0(x) = 0, \quad \phi_0(x) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{-ipx} + a_p^\dagger e^{ipx})$$

Hamiltonian and one particle state

$$H_0 = \int \frac{d^3 p}{(2\pi)^3} \omega_p \left(a_p^\dagger a_p + \frac{1}{2} \right), \quad H_0 |0\rangle = E_0 |0\rangle$$
$$a_p |0\rangle = 0, \quad a_p^\dagger |0\rangle = \frac{1}{\sqrt{2\omega_p}} |p\rangle$$

Recap

QFT study#1 :

Interaction Theory

Lagrangian

$$\mathcal{L} = \mathcal{L}_{\text{kinetic}} + \mathcal{L}_{\text{int}} = -\frac{1}{2}\phi(\square + m^2)\phi + \mathcal{L}_{\text{int}}[\phi]$$

EOM and free field

$$(\square + m^2)\phi(x) = \mathcal{L}'_{\text{int}}[\phi], \quad \phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p(t)e^{-ipx} + a_p^\dagger(t)e^{ipx})$$

Hamiltonian and one particle state

$$H = H_0 + V, \quad V = - \int d^3x \mathcal{L}_{\text{int}}[\phi]$$
$$H|\Omega\rangle = E_{\text{ground}}|\Omega\rangle$$
$$a_p^\dagger(\pm\infty)|\Omega\rangle = \frac{1}{\sqrt{2\omega_p}}|p; \pm\infty\rangle, \quad a_p(\pm\infty)|\Omega\rangle = 0$$

Recap

QFT study#2 :

Scattering cross section

Differential cross section

$$d\sigma = \frac{1}{2E_1 2E_2 v_{moll}} |\mathcal{M}|^2 d\Pi_{\text{LIPS}}$$

Especially 2 to 2 in CM frame

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{CM}} = \frac{1}{64\pi^2 s} \frac{|\vec{p}_f|}{|\vec{p}_i|} |\mathcal{M}|^2 \theta(\sqrt{s} - m_3 - m_4)$$

Recap

QFT study#2 :

Scattering cross section

Matrix element $\mathcal{M}$ from

$$\langle f|S - 1|i\rangle = i(2\pi)^4 \delta^4(\Sigma p_{ini} - \Sigma p_{fin}) \langle f|\mathcal{M}|i\rangle$$

To find $\mathcal{M}$, let's consider $\langle f|S - 1|i\rangle$

LSZ reduction formula

$$\langle f|S|i\rangle = \langle f; \infty|i; -\infty\rangle$$

In 2 to n process, $|i; -\infty\rangle = |p_1; -\infty, p_2; -\infty\rangle$, $|f; \infty\rangle = |p_3; \infty, p_4; \infty, \dots, p_n; \infty\rangle$

$$|i; -\infty\rangle = \sqrt{2\omega_{p_1} 2\omega_{p_2}} a_{p_1}^\dagger(-\infty) a_{p_2}^\dagger(-\infty) |\Omega\rangle$$
$$|f; \infty\rangle = \sqrt{2\omega_{p_3} \cdots 2\omega_{p_n}} a_{p_3}^\dagger(\infty) \cdots a_{p_n}^\dagger(\infty) |\Omega\rangle$$

So,

$$\langle f|S|i\rangle = \sqrt{2\omega_{p_1} \cdots 2\omega_{p_n}} \langle \Omega | a_{p_3}(\infty) \cdots a_{p_n}(\infty) a_{p_1}^\dagger(-\infty) a_{p_2}^\dagger(-\infty) | \Omega \rangle$$

This expression is not terribly useful as is.

LSZ reduction formula

$$\begin{aligned}\langle f|S|i\rangle &= \sqrt{2\omega_{p_1} \dots 2\omega_{p_n}} \langle \Omega | a_{p_3}(\infty) \dots a_{p_n}(\infty) a_{p_1}^\dagger(-\infty) a_{p_2}^\dagger(-\infty) | \Omega \rangle \\ &= \sqrt{2\omega_{p_1} \dots 2\omega_{p_n}} \langle \Omega | T \left\{ \left(a_{p_3}(\infty) - a_{p_3}(-\infty) \right) \dots \left(a_{p_2}^\dagger(-\infty) - a_{p_2}^\dagger(\infty) \right) \right\} | \Omega \rangle\end{aligned}$$

$T\{\dots\}$ is Time ordering operator.

Unwanted annihilation operators (like $a_{p_3}(-\infty)$) are ordered by T , they go to right and annihilate vacuum which give zero.

Similarly, creation operators go left and annihilate bra-state vacuum.

LSZ reduction formula

Using

$$i \int d^4x e^{ipx} (\square + m^2) \phi(x) = \sqrt{2\omega_p} [a_p(\infty) - a_p(-\infty)]$$

And

$$i \int d^4x e^{-ipx} (\square + m^2) \phi(x) = \sqrt{2\omega_p} [a_p^\dagger(-\infty) - a_p^\dagger(\infty)]$$

Derivation for appendix

LSZ reduction formula

$$\begin{aligned}
 \langle f|S|i\rangle &= \sqrt{2\omega_{p_1} \dots 2\omega_{p_n}} \langle \Omega | a_{p_3}(\infty) \dots a_{p_n}(\infty) a_{p_1}^\dagger(-\infty) a_{p_2}^\dagger(-\infty) | \Omega \rangle \\
 &= \sqrt{2\omega_{p_1} \dots 2\omega_{p_n}} \langle \Omega | T \left\{ \left(a_{p_3}(\infty) - a_{p_3}(-\infty) \right) \dots \left(a_{p_2}^\dagger(-\infty) - a_{p_2}^\dagger(\infty) \right) \right\} | \Omega \rangle \\
 &= \left(i \int d^4x_1 e^{-ip_1x_1} (\square_1 + m^2) \dots i \int d^4x_n e^{ip_nx_n} (\square_n + m^2) \right) \\
 &\quad \langle \Omega | T \{ \phi(x_1) \phi(x_2) \phi(x_3) \dots \phi(x_n) \} | \Omega \rangle
 \end{aligned}$$

LSZ reduction formula

$$\begin{aligned}
 \langle f|S|i\rangle &= \left(i \int d^4x_1 e^{-ip_1x_1} (\square_1 + m^2) \dots i \int d^4x_n e^{ip_nx_n} (\square_n + m^2) \right) \\
 &\quad \langle \Omega | T \{ \phi(x_1) \phi(x_2) \phi(x_3) \dots \phi(x_n) \} | \Omega \rangle
 \end{aligned}$$

LSZ reduction formula

$$\langle f|S|i\rangle = \left(i \int d^4x_1 e^{-ip_1x_1} (\square_1 + m^2) \cdots i \int d^4x_n e^{ip_nx_n} (\square_n + m^2) \right) \langle \Omega|T\{\phi(x_1)\phi(x_2)\phi(x_3)\cdots\phi(x_n)\}|\Omega\rangle$$

IF ϕ is free field, $(\square + m^2)\phi = 0$, so $(\square + m^2)$ terms give zero.

But, as we will see, $\langle \Omega| \dots |\Omega\rangle$ part gives us propagators $\frac{1}{\square + m^2}$ for the one particle state, these blow up. LSZ cancel them. Especially external particles are only canceled.

Another fields which we don't want to make them external particles will be killed.

Feynman propagator

To calculate $\langle f|S|i\rangle$, we need to compute Time ordered products of fields. As an example, let's calculate 2 point function in free theory.

$$\langle 0|T\{\phi_0(x_1)\phi_0(x_2)\}|0\rangle$$

We can compute $\langle 0|\phi_0(x_1)\phi_0(x_2)|0\rangle$ easily,

$$\langle 0|\phi_0(x_1)\phi_0(x_2)|0\rangle = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{ip(x_2-x_1)}$$

Feynman propagator

$$\begin{aligned}\langle 0|T\{\phi_0(x_1)\phi_0(x_2)\}|0\rangle &= \langle 0|\phi_0(x_1)\phi_0(x_2)|0\rangle\theta(t_1 - t_2) + \langle 0|\phi_0(x_2)\phi_0(x_1)|0\rangle\theta(t_2 - t_1) \\&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{ip(x_2-x_1)}\theta(t_1 - t_2) + \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{ip(x_1-x_2)}\theta(t_2 - t_1) \\&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{i\omega_p(t_2-t_1)} e^{i\vec{p}(\vec{x}_1-\vec{x}_2)}\theta(t_1 - t_2) \\&\quad + \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{i\omega_p(t_1-t_2)} e^{-i\vec{p}(\vec{x}_1-\vec{x}_2)}\theta(t_2 - t_1) \\&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{-i\vec{p}(\vec{x}_1-\vec{x}_2)} \left(e^{-i\omega_p\tau}\theta(\tau) + e^{i\omega_p\tau}\theta(-\tau) \right)\end{aligned}$$

Feynman propagator

Mathematical identity

$$e^{-i\omega_p\tau}\theta(\tau) = \frac{1}{2\pi i} \int_{-\infty}^{\infty} d\omega \frac{e^{i\omega\tau}}{\omega - (-\omega_p + i\epsilon)}$$

and

$$e^{i\omega_p\tau}\theta(-\tau) = \frac{-1}{2\pi i} \int_{-\infty}^{\infty} d\omega \frac{e^{i\omega\tau}}{\omega - (\omega_p - i\epsilon)}$$

Feynman propagator

$$\begin{aligned}\langle 0|T\{\phi_0(x_1)\phi_0(x_2)\}|0\rangle &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{-i\vec{p}(\vec{x}_1-\vec{x}_2)} \\ &\quad \times \left(\frac{1}{2\pi i} \int_{-\infty}^{\infty} d\omega \frac{e^{i\omega\tau}}{\omega - (-\omega_p + i\epsilon)} - \frac{1}{2\pi i} \int_{-\infty}^{\infty} d\omega \frac{e^{i\omega\tau}}{\omega - (\omega_p - i\epsilon)} \right) \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{-i\vec{p}(\vec{x}_1-\vec{x}_2)} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{i\omega(t_1-t_2)} \frac{2\omega_p i}{\omega^2 - \omega_p^2 + i\epsilon} \\ &= \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{ip(x_1-x_2)}\end{aligned}$$

$$\langle 0|T\{\phi_0(x_1)\phi_0(x_2)\}|0\rangle = \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{ip(x_1-x_2)}$$

Now, $p^\mu = (\omega, \vec{p})$, $p^0 = \omega \neq \sqrt{\vec{p}^2 + m^2}$! Propagating field can be off-shell

Feynman propagator

$$\begin{aligned}\langle 0|T\{\phi_0(x_1)\phi_0(x_2)\}|0\rangle &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{-i\vec{p}(\vec{x}_1-\vec{x}_2)} \\ &\quad \times \left(\frac{1}{2\pi i} \int_{-\infty}^{\infty} d\omega \frac{e^{i\omega\tau}}{\omega - (-\omega_p + i\epsilon)} - \frac{1}{2\pi i} \int_{-\infty}^{\infty} d\omega \frac{e^{i\omega\tau}}{\omega - (\omega_p - i\epsilon)} \right) \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{-i\vec{p}(\vec{x}_1-\vec{x}_2)} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{i\omega(t_1-t_2)} \frac{2\omega_p i}{\omega^2 - \omega_p^2 + i\epsilon} \\ &= \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{ip(x_1-x_2)}\end{aligned}$$

$$\langle 0|T\{\phi_0(x_1)\phi_0(x_2)\}|0\rangle = \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{ip(x_1-x_2)}$$

For $\epsilon = 0$ the Feynman propagator looks just like a classical Green's function for the Klein-Gordon equation $(\square + m^2) D_F(x, y) = -i\delta^4(x)$ with certain boundary conditions. That is because it is. We are just computing classical propagation in a really complicated way.

Interaction picture

What we have to calculate is

$$\langle f|S|i\rangle \sim \text{LSZ} \langle \Omega | T \{ \phi(x_1) \phi(x_2) \dots \phi(x_n) \} | \Omega \rangle$$

$\phi(x)$ is Heisenberg picture field, we don't know well, and $|\Omega\rangle$ is interaction theory vacuum, $H|\Omega\rangle = E_{\text{ground}}|\Omega\rangle$, again we don't know well.

We are EXPERT at free theory. And already know time-ordered products of field in free theory – Feynman propagator. How can we treat them...

Interaction picture – Field

The main idea is this : we know free fields, but we have no idea what the Heisenberg field is doing. So at some reference time t_0 we can match the free field and the Heisenberg field and from then on we evolve our field only with H_0 . Congratulations — for the rest of the calculation, it is a free field forever.

$$\begin{aligned}\phi(x) &= S^\dagger(t, t_0)\phi(\vec{x}, t_0)S(t, t_0) \\ &= S^\dagger(t, t_0)\phi_0(\vec{x}, t_0)S(t, t_0) \\ &= S^\dagger(t, t_0)e^{-iH_0(t-t_0)}e^{iH_0(t-t_0)}\phi_0(\vec{x}, t_0)e^{-iH_0(t-t_0)}e^{iH_0(t-t_0)}S(t, t_0) \\ &= U^\dagger(t, t_0)\phi_0(x)U(t, t_0)\end{aligned}$$

where $i\partial_t S(t, t_0) = S(t, t_0)H(t) = H(t_0)S(t, t_0)$ and $U(t, t_0) = e^{iH_0(t-t_0)}S(t, t_0)$

Interaction picture – Field

Interacting time evolution part encode in $U(t, t_0)$

And we can calculate ; $i\partial_t U(t, t_0) = V_I(t)U(t, t_0)$ where $V_I(t) = e^{iH_0(t-t_0)}V(t_0)e^{-iH_0(t-t_0)}$

$$\begin{aligned} i\partial_t U(t, t_0) &= -e^{iH_0(t-t_0)}H_0S(t, t_0) + e^{iH_0(t-t_0)}H(t_0)S(t, t_0) \\ &= e^{iH_0(t-t_0)}[-H_0 + H(t_0)]e^{-iH_0(t-t_0)}e^{iH_0(t-t_0)}S(t, t_0) \\ &= e^{iH_0(t-t_0)}V(t_0)e^{-iH_0(t-t_0)}U \\ &= V_I(t)U(t, t_0) \end{aligned}$$

where $V(t) = -\int d^3x \mathcal{L}_{\text{int}}[\phi(x)]$

Interaction picture – Field

$$\begin{aligned} V(t_0) &= - \int d^3x \mathcal{L}_{\text{int}}[\phi(\vec{x}, t_0)] \\ &= - \int d^3x \mathcal{L}_{\text{int}}[\phi_0(\vec{x}, t_0)] \end{aligned}$$

therefore

$$\begin{aligned} V_I(t) &= e^{iH_0(t-t_0)} \left(- \int d^3x \mathcal{L}_{\text{int}}[\phi_0(\vec{x}, t_0)] \right) e^{-iH_0(t-t_0)} \\ &= - \int d^3x \mathcal{L}_{\text{int}}[\phi_0(x)] \end{aligned}$$

Interaction picture – Field

As the same way to get $S = \bar{T} \exp(-i \int dt H)$ from $i \partial_t S = SH$, we can get $U(t, t_0)$ from $i \partial_t U = V_I U$

$$\begin{aligned} U(t, t_0) &= 1 - i \int_{t_0}^t dt' V(t') U(t', t_0) \\ &= 1 - i \int_{t_0}^t dt' V(t') + (-i)^2 \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' V(t') V(t'') + \dots \\ &= 1 - i \int_{t_0}^t dt' V(t') + \frac{(-i)^2}{2!} \int_{t_0}^t dt' \int_{t_0}^t dt'' T\{V(t') V(t'')\} + \dots \\ &= T \exp \left(-i \int_{t_0}^t dt' V_I(t') \right) = T \exp \left(i \int_{t_0}^t dt' \int d^3x \mathcal{L}_{\text{int}}[\phi_0] \right) \end{aligned}$$

Interaction picture – Field

When we denote

$$U_{21} = U(t_2, t_1) = T \exp \left(-i \int_{t_1}^{t_2} dt V_I(t) \right) = T \exp \left(i \int_{t_1}^{t_2} d^4x \mathcal{L}_{\text{int}}[\phi_0] \right)$$

We can easily check

$$U_{12}U_{21} = 1, \quad U_{12} = U_{21}^{-1} = U_{21}^\dagger$$

For ANY time ordering,

$$U_{32}U_{21} = U_{31}, \quad U_{31}U_{12} = U_{32}, \quad U_{13}U_{32} = U_{12}$$

$$\text{So, } \phi(x_1) = U_{10}^\dagger \phi_0(x_1) U_{10} = U_{01} \phi_0(x_1) U_{10}$$

Interaction picture – Vacuum

The main idea is this : at asymptotic times, we assume that the interacting vacuum contains zero asymptotic particles. In other words, the asymptotic annihilation operators annihilate the interacting vacuum.

$$a_p(t = \pm\infty)|\Omega\rangle = 0$$

Using the time-evolution relation for the annihilation operator, this means that the time-evolved interacting vacuum is annihilated by every free annihilation operator. But the only state with that property is the free vacuum!

$$U^\dagger(\pm\infty, t_0)a_p U(\pm\infty, t_0)|\Omega\rangle = 0 \rightarrow a_p U(\pm\infty, t_0)|\Omega\rangle = 0$$

$$(\therefore) U(\pm\infty, t_0)|\Omega\rangle = \mathcal{N}|0\rangle$$

Interaction picture – Vacuum

$$|\Omega\rangle = \mathcal{N}_i U_{0-\infty} |0\rangle, \quad \langle\Omega| = \mathcal{N}_f \langle 0| U_{\infty 0}$$

With Normalization condition $\langle\Omega|\Omega\rangle = 1$ gives

$$\mathcal{N}_i \mathcal{N}_f = \frac{1}{\langle 0| U_{\infty -\infty} |0\rangle}$$

Therefore, by a beautifully shameless trick, we have managed to turn both the vacuum and the field into free-theory objects.

Interaction picture – Assemble!

$$\langle \Omega | T \{ \phi(x_1) \cdots \phi(x_n) \} | \Omega \rangle$$

We know about fields $\Rightarrow \phi(x_j) = U_{0j} \phi_0(x_j) U_{j0}$

And we also know vacuum ! $\Rightarrow |\Omega\rangle = \mathcal{N}_i U_{0-\infty} |0\rangle$, $\langle \Omega| = \mathcal{N}_f \langle 0| U_{\infty 0}$

with normalization factor $\mathcal{N}_i \mathcal{N}_f = \frac{1}{\langle 0| U_{\infty -\infty} |0\rangle}$

$$\begin{aligned} \langle \Omega | T \{ \phi(x_1) \cdots \phi(x_n) \} | \Omega \rangle &= \frac{\langle 0 | T \{ U_{\infty 0} U_{01} \phi_0(x_1) U_{10} U_{02} \phi_0(x_2) U_{20} \cdots U_{0n} \phi_0(x_n) U_{n0} U_{0-\infty} \} | 0 \rangle}{\langle 0 | U_{\infty -\infty} | 0 \rangle} \\ &= \frac{\langle 0 | T \{ U_{\infty 1} \phi_0(x_1) U_{12} \phi_0(x_2) U_{23} \phi_0(x_3) \cdots \phi_0(x_{n-1}) U_{n-1,n} \phi_0(x_n) U_{n-\infty} \} | 0 \rangle}{\langle 0 | U_{\infty -\infty} | 0 \rangle} \end{aligned}$$

Interaction picture – Assemble!

$$\begin{aligned}\langle\Omega|T\{\phi(x_1)\cdots\phi(x_n)\}|\Omega\rangle &= \frac{\langle 0|T\{U_{\infty 1}\phi_0(x_1)U_{12}\phi_0(x_2)U_{23}\phi_0(x_3)\cdots\phi_0(x_{n-1})U_{n-1,n}\phi_0(x_n)U_{n-\infty}\}|0\rangle}{\langle 0|U_{\infty-\infty}|0\rangle} \\ &= \frac{\langle 0|T\{\phi_0(x_1)\phi_0(x_2)\cdots\phi_0(x_n)U_{\infty-\infty}\}|0\rangle}{\langle 0|U_{\infty-\infty}|0\rangle}\end{aligned}$$

$$\langle\Omega|T\{\phi(x_1)\cdots\phi(x_n)\}|\Omega\rangle = \frac{\langle 0|T\{\phi_0(x_1)\phi_0(x_2)\cdots\phi_0(x_n)\exp(i\int d^4x\mathcal{L}_{\text{int}})\}|0\rangle}{\langle 0|T\exp(i\int d^4x\mathcal{L}_{\text{int}})|0\rangle}$$

When we solve the problem, we expand $\exp(i\int d^4x\mathcal{L}_{\text{int}}) = 1 + i\int d^4x\mathcal{L}_{\text{int}} + \cdots$

Contraction

Let's take an example ϕ^3 theory, $\mathcal{L}_{\text{int}} = \frac{g}{3!} \phi^3$ and consider $\langle \Omega | T \{ \phi(x_1) \phi(x_2) \} | \Omega \rangle$

$$\langle \Omega | T \{ \phi(x_1) \phi(x_2) \} | \Omega \rangle = \frac{\langle 0 | T \left\{ \phi_0(x_1) \phi_0(x_2) \exp \left(i \int d^4x \frac{g}{3!} \phi_0(x)^3 \right) \right\} | 0 \rangle}{\langle 0 | T \exp \left(i \int d^4x \frac{g}{3!} \phi_0(x)^3 \right) | 0 \rangle}$$

let's expand exponential part in numerator perturbatively in g

$$\begin{aligned} & \langle 0 | T \left\{ \phi_0(x_1) \phi_0(x_2) \exp \left(i \int d^4x \frac{g}{3!} \phi_0(x)^3 \right) \right\} | 0 \rangle \\ &= \langle 0 | T \phi_0(x_1) \phi_0(x_2) | 0 \rangle + \frac{ig}{3!} \int d^4x \langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 | 0 \rangle \\ & \quad + \frac{1}{2} \left(\frac{ig}{3!} \right)^2 \int d^4x \int d^4y \langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 \phi_0(y)^3 | 0 \rangle \end{aligned}$$

Contraction

$$\begin{aligned}
 & \langle 0 | T \left\{ \phi_0(x_1) \phi_0(x_2) \exp \left(i \int d^4x \frac{g}{3!} \phi_0(x)^3 \right) \right\} | 0 \rangle \\
 &= \langle 0 | T \phi_0(x_1) \phi_0(x_2) | 0 \rangle + \frac{ig}{3!} \int d^4x \langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 | 0 \rangle \\
 & \quad + \frac{1}{2} \left(\frac{ig}{3!} \right)^2 \int d^4x \int d^4y \langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 \phi_0(y)^3 | 0 \rangle
 \end{aligned}$$

Creation and annihilation must be matched! For example, in the circled part If $t_1 > t_2 > x^0$, then

$$\langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 | 0 \rangle = \langle 0 | \phi_0(x_1) \phi_0(x_2) \phi_0(x) \phi_0(x) \phi_0(x) | 0 \rangle$$

a_p
 $a_p^\dagger$

Contraction

$$\begin{aligned}
 & \langle 0 | T \left\{ \phi_0(x_1) \phi_0(x_2) \exp \left(i \int d^4x \frac{g}{3!} \phi_0(x)^3 \right) \right\} | 0 \rangle \\
 &= \langle 0 | T \phi_0(x_1) \phi_0(x_2) | 0 \rangle + \frac{ig}{3!} \int d^4x \langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 | 0 \rangle \\
 & \quad + \frac{1}{2} \left(\frac{ig}{3!} \right)^2 \int d^4x \int d^4y \langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 \phi_0(y)^3 | 0 \rangle
 \end{aligned}$$

Creation and annihilation must be matched! For example, in the circled part If $t_1 > t_2 > x^0$, then

$$\langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 | 0 \rangle = \langle 0 | \phi_0(x_1) \phi_0(x_2) \phi_0(x) \phi_0(x) \phi_0(x) | 0 \rangle$$

The diagram illustrates the contraction of the expression. The first two fields are blue and the last three are red. Blue arcs connect the first blue field to the first red field (labeled a_p) and the second blue field to the second red field (labeled a_k). Red arcs connect the third and fourth red fields (labeled $a_k^\dagger$) and the fifth red field to the second blue field (labeled $a_p^\dagger$).

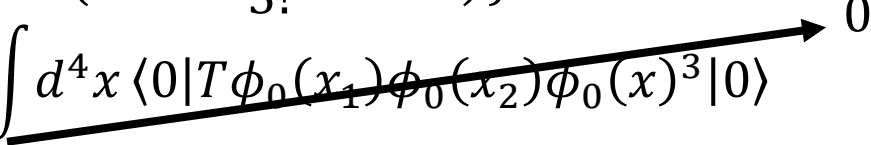
Contraction

$$\begin{aligned}
 & \langle 0 | T \left\{ \phi_0(x_1) \phi_0(x_2) \exp \left(i \int d^4x \frac{g}{3!} \phi_0(x)^3 \right) \right\} | 0 \rangle \\
 &= \langle 0 | T \phi_0(x_1) \phi_0(x_2) | 0 \rangle + \frac{ig}{3!} \int d^4x \langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 | 0 \rangle \\
 & \quad + \frac{1}{2} \left(\frac{ig}{3!} \right)^2 \int d^4x \int d^4y \langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 \phi_0(y)^3 | 0 \rangle
 \end{aligned}$$

Creation and annihilation must be matched! For example, in the circled part If $t_1 > t_2 > x^0$, then

$$\langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 | 0 \rangle = \langle 0 | \phi_0(x_1) \phi_0(x_2) \phi_0(x) \phi_0(x) \phi_0(x) | 0 \rangle = 0$$

Contraction

$$\begin{aligned}
 & \langle 0 | T \left\{ \phi_0(x_1) \phi_0(x_2) \exp \left(i \int d^4x \frac{g}{3!} \phi_0(x)^3 \right) \right\} | 0 \rangle \\
 &= \langle 0 | T \phi_0(x_1) \phi_0(x_2) | 0 \rangle + \frac{ig}{3!} \int d^4x \langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 | 0 \rangle \\
 &\quad + \frac{1}{2} \left(\frac{ig}{3!} \right)^2 \int d^4x \int d^4y \langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 \phi_0(y)^3 | 0 \rangle
 \end{aligned}$$


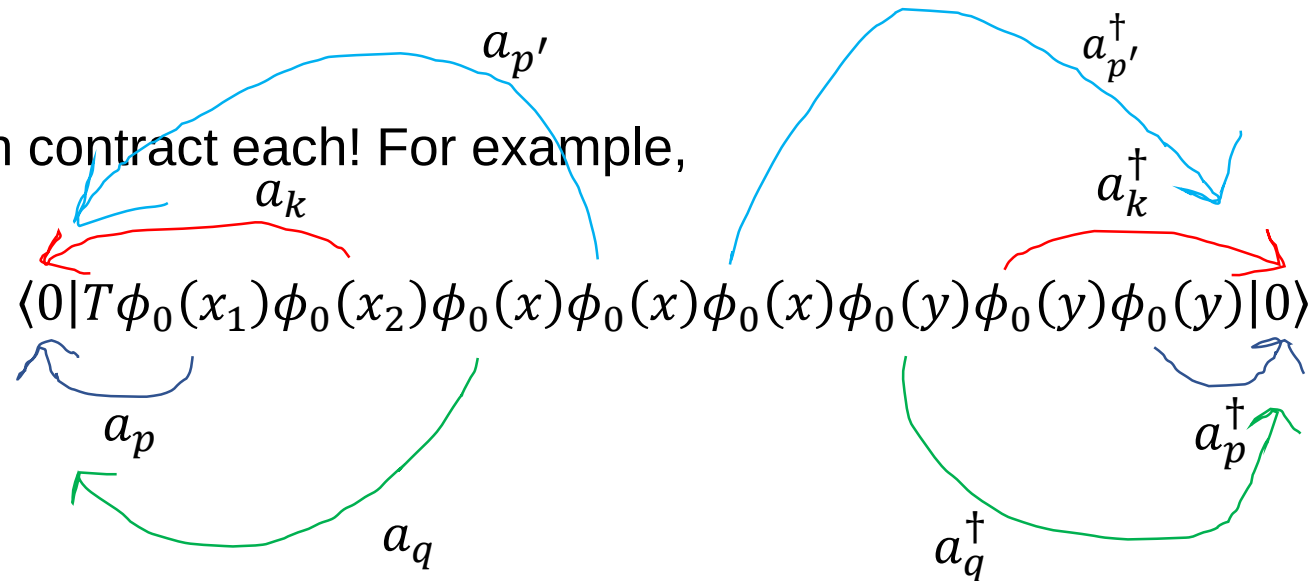
g^2 order term; can contract each! For example,

$$\langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x) \phi_0(x) \phi_0(x) \phi_0(y) \phi_0(y) \phi_0(y) | 0 \rangle$$

Contraction

$$\begin{aligned}
 & \langle 0 | T \left\{ \phi_0(x_1) \phi_0(x_2) \exp \left(i \int d^4x \frac{g}{3!} \phi_0(x)^3 \right) \right\} | 0 \rangle \\
 &= \langle 0 | T \phi_0(x_1) \phi_0(x_2) | 0 \rangle + \frac{ig}{3!} \int d^4x \langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 | 0 \rangle \\
 & \quad + \frac{1}{2} \left(\frac{ig}{3!} \right)^2 \int d^4x \int d^4y \langle 0 | T \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 \phi_0(y)^3 | 0 \rangle
 \end{aligned}$$

g^2 order term; can contract each! For example,



Contraction

$$\begin{aligned} & \langle 0|T \left\{ \phi_0(x_1)\phi_0(x_2) \exp \left(i \int d^4x \frac{g}{3!} \phi_0(x)^3 \right) \right\} |0\rangle \\ &= \langle 0|T \phi_0(x_1)\phi_0(x_2)|0\rangle + \frac{1}{2} \left(\frac{ig}{3!} \right)^2 \int d^4x \int d^4y \langle 0|T \phi_0(x_1)\phi_0(x_2)\phi_0(x)^3\phi_0(y)^3|0\rangle \end{aligned}$$

Wick theorem guarantees ; each contraction gives “Two point function” – yes. Exactly Feynman propagator. Let $\langle 0|T \phi_0(x_1)\phi_0(x_2)|0\rangle = D_{12}$

$$\begin{aligned} & \langle 0|T \phi_0(x_1)\phi_0(x_2)\phi_0(x)\phi_0(x)\phi_0(x)\phi_0(y)\phi_0(y)\phi_0(y)|0\rangle \\ &= 9D_{12}D_{xx}D_{xy}D_{yy} + 6D_{12}D_{xy}^3 + 18D_{1x}D_{2x}D_{xy}D_{yy} + 9D_{1x}D_{2y}D_{xx}D_{yy} + 18D_{1x}D_{2y}D_{xy}^2 \\ & \quad + 18D_{1y}D_{2y}D_{xy}D_{xx} + 9D_{1y}D_{2x}D_{xx}D_{yy} + 18D_{1y}D_{2x}D_{xy}^2 \end{aligned}$$

Position space Feynman rules

So,

$$\begin{aligned}
 & \langle \Omega | T \{ \phi(x_1) \phi(x_2) \} | \Omega \rangle \\
 &= \frac{1}{\langle 0 | T \exp \left(i \int d^4x \frac{g}{3!} \phi_0(x)^3 \right) | 0 \rangle} \left(D_{12} \right. \\
 & \quad \left. - g^2 \int d^4x \int d^4y \left(\frac{1}{8} D_{12} D_{xx} D_{xy} D_{yy} + \frac{1}{12} D_{12} D_{xy}^3 + \frac{1}{2} D_{1x} D_{2x} D_{xy} D_{yy} + \frac{1}{4} D_{1x} D_{2y} D_{xx} D_{yy} + \frac{1}{2} D_{1x} D_{2y} D_{xy}^2 \right) \right) \\
 &=
 \end{aligned}$$

Position space Feynman rule

So,

$$\begin{aligned}
 & \langle \Omega | T \{ \phi(x_1) \phi(x_2) \} | \Omega \rangle \\
 &= \frac{1}{\langle 0 | T \exp \left(i \int d^4x \frac{g}{3!} \phi_0(x)^3 \right) | 0 \rangle} \left(D_{12} \right. \\
 & \quad \left. - g^2 \int d^4x \int d^4y \left(\frac{1}{8} D_{12} D_{xx} D_{xy} D_{yy} + \frac{1}{12} D_{12} D_{xy}^3 + \frac{1}{2} D_{1x} D_{2x} D_{xy} D_{yy} + \frac{1}{4} D_{1x} D_{2y} D_{xx} D_{yy} + \frac{1}{2} D_{1x} D_{2y} D_{xy}^2 \right) \right) \\
 &=
 \end{aligned}$$

1. External points

각 x_i 에 external point를 하나 그린다.

2. Propagator

두 점 x, y 를 연결하는 선 하나마다

$$D_F(x - y)$$

3. Interaction vertex

$\mathcal{L}_{\text{int}}$ 의 interaction 하나마다 새로운 internal vertex z 를 만든다.

각 vertex에는 **Lagrangian**의 **interaction coefficient**를 붙인다.

4. Integrate internal vertices

$$\int d^4z$$

모든 internal vertex의 위치를 적분한다.

5. Draw all possible diagrams

원하는 perturbation order에서 가능한 모든 contraction 구조를 그리고 전부 더한다.

6. No vacuum bubbles

external point와 전혀 연결되지 않은 subdiagram은 제외한다.

7. Symmetry factor

동일한 diagram을 만드는 중복이 있으면 geometrical symmetry factor로 나눈다.

Position space Feynman rules

Denominator in order of g^2 gives

$$\langle 0|T \exp\left(i \int d^4x \frac{g}{3!} \phi_0(x)^3\right) |0\rangle = 1 + \left(\frac{ig}{3!}\right)^2 \frac{1}{2} \int d^4x \int d^4y (9D_{xx}D_{xy}D_{yy} + 6D_{xy}^3) + \mathcal{O}(g^3)$$

Since numerator

$$D_{12} + \left(\frac{ig}{3!}\right)^2 \frac{1}{2} g^2 \int d^4x \int d^4y \left(D_{12}(9D_{xx}D_{xy}D_{yy} + 6D_{xy}^3) + \frac{1}{2} D_{1x}D_{2x}D_{xy}D_{yy} + \frac{1}{4} D_{1x}D_{2y}D_{xx}D_{yy} + \frac{1}{2} D_{1x}D_{2y}D_{xy}^2 \right)$$

Position space Feynman rules

Denominator in order of g^2 gives

$$\langle 0|T \exp\left(i \int d^4x \frac{g}{3!} \phi_0(x)^3\right) |0\rangle = 1 + \left(\frac{ig}{3!}\right)^2 \frac{1}{2} \int d^4x \int d^4y \overbrace{(9D_{xx}D_{xy}D_{yy} + 6D_{xy}^3)}^{\text{bubble}} + \mathcal{O}(g^3)$$

Since numerator

$$D_{12} + \left(\frac{ig}{3!}\right)^2 \frac{1}{2} g^2 \int d^4x \int d^4y \left(\overbrace{D_{12}(9D_{xx}D_{xy}D_{yy} + 6D_{xy}^3)}^{\text{bubble}} + \frac{1}{2} D_{1x}D_{2x}D_{xy}D_{yy} + \frac{1}{4} D_{1x}D_{2y}D_{xx}D_{yy} + \frac{1}{2} D_{1x}D_{2y}D_{xy}^2 \right)$$

Position space Feynman rules

$$\begin{aligned}
 \langle \Omega | T \{ \phi(x_1) \phi(x_2) \} | \Omega \rangle &= \frac{D_{12}(1 + g^2[\text{Bubble}]) + g^2[\text{no bubble}]}{1 + g^2(\text{Bubble})} \\
 &= [D_{12}(1 + g^2[\text{Bubble}]) + g^2[\text{no bubble}]] [1 - g^2(\text{Bubble})] \\
 &= D_{12} + g^2[\text{no bubble}] + \mathcal{O}(g^4)
 \end{aligned}$$

Generally, bubbles are always canceled. In diagrams,

$$\begin{aligned}
 & \text{---} + \text{---} \bigcirc \text{---} + \frac{\text{---} \bigcirc \text{---}}{\text{---}} + \frac{\text{---} \bigcirc \text{---}}{\text{---}} + \dots \\
 &= \left(\text{---} + \text{---} \bigcirc \text{---} + \dots \right) \times \left(1 + \text{---} \bigcirc \text{---} + \text{---} \bigcirc \text{---} + \dots \right).
 \end{aligned}
 \tag{7.76}$$

Position space Feynman rules

So,

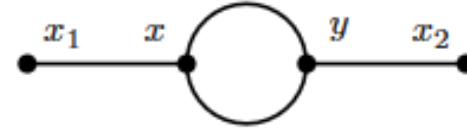
$$\langle \Omega | T \{ \phi(x_1) \phi(x_2) \} | \Omega \rangle = \langle 0 | T \{ \phi_0(x_1) \phi_0(x_2) e^{i \int \mathcal{L}_{\text{int}}} \} | 0 \rangle_{\text{no bubbles}}$$

Stepping back and looking at the overall picture,

$$\begin{aligned} \langle p_2 | S | p_1 \rangle &= i \int d^4 x_1 e^{-i p_1 x_1} (\square_1 + m^2) i \int d^4 x_2 e^{i p_2 x_2} (\square_2 + m^2) \langle \Omega | T \{ \phi(x_1) \phi(x_2) \} | \Omega \rangle \\ &= i \int d^4 x_1 e^{-i p_1 x_1} (\square_1 + m^2) i \int d^4 x_2 e^{i p_2 x_2} (\square_2 + m^2) \langle 0 | T \{ \phi_0(x_1) \phi_0(x_2) e^{i \int \mathcal{L}_{\text{int}}} \} | 0 \rangle_{\text{no bubbles}} \end{aligned}$$

Momentum space Feynman rules

Let's think order g^2 connected and no bubbles diagram,



$$\begin{aligned}
 & \langle 0 | T \{ \phi_0(x_1) \phi_0(x_2) e^{i \int \mathcal{L}_{\text{int}}} \} | 0 \rangle_{\text{no bubbles}} \\
 &= \frac{(ig)^2}{2} \int d^4x \int d^4y D_{1x} D_{xy}^2 D_{y2} \\
 &= \frac{(ig)^2}{2} \int d^4x d^4y \int \frac{d^4k_1}{(2\pi)^4} \frac{d^4q_1}{(2\pi)^4} \frac{d^4q_2}{(2\pi)^4} \frac{d^4k_2}{(2\pi)^4} \frac{ie^{ik_1(x_1-x)}}{k_1^2 - m^2 + i\epsilon} \frac{ie^{iq_1(x-y)}}{q_1^2 - m^2 + i\epsilon} \frac{ie^{iq_2(x-y)}}{q_2^2 - m^2 + i\epsilon} \frac{ie^{ik_2(y-x_2)}}{k_2^2 - m^2 + i\epsilon} \\
 &= \frac{(ig)^2}{2} \int \frac{d^4k_1}{(2\pi)^4} \frac{d^4q_1}{(2\pi)^4} \frac{d^4k_2}{(2\pi)^4} \frac{i}{k_1^2 - m^2 + i\epsilon} \frac{i}{q_1^2 - m^2 + i\epsilon} \frac{i}{(k_1 - q_1)^2 - m^2 + i\epsilon} \frac{i}{k_2^2 - m^2 + i\epsilon} e^{ik_1x_1} e^{-ik_2x_2} (2\pi)^4 \delta^4(k_1 - k_2)
 \end{aligned}$$

Momentum space Feynman rules

$$\langle p_2 | S | p_1 \rangle = i \int d^4 x_1 e^{-ip_1 x_1} (\square_1 + m^2) i \int d^4 x_2 e^{ip_2 x_2} (\square_2 + m^2) \langle 0 | T \{ \phi_0(x_1) \phi_0(x_2) e^{i \int \mathcal{L}_{\text{int}}} \} | 0 \rangle_{\text{no bubbles}}$$

$$\begin{aligned} \langle p_2 | S - 1 | p_1 \rangle &= i \int d^4 x_1 e^{-ip_1 x_1} (\square_1 + m^2) i \int d^4 x_2 e^{ip_2 x_2} (\square_2 + m^2) \frac{(ig)^2}{2} \int \frac{d^4 k_1}{(2\pi)^4} \frac{d^4 q_1}{(2\pi)^4} \frac{d^4 k_2}{(2\pi)^4} \frac{i}{k_1^2 - m^2 + i\epsilon} \\ &\quad \times \frac{i}{q_1^2 - m^2 + i\epsilon} \frac{i}{(k_1 - q_1)^2 - m^2 + i\epsilon} \frac{i}{k_2^2 - m^2 + i\epsilon} e^{ik_1 x_1} e^{-ik_2 x_2} (2\pi)^4 \delta^4(k_1 - k_2) \\ &= i \int d^4 x_1 e^{-ip_1 x_1} \cancel{(-k_1^2 + m^2)} i \int d^4 x_2 e^{ip_2 x_2} \cancel{(-k_2^2 + m^2)} \frac{(ig)^2}{2} \int \frac{d^4 k_1}{(2\pi)^4} \frac{d^4 q_1}{(2\pi)^4} \frac{d^4 k_2}{(2\pi)^4} \cancel{k_1^2 - m^2 + i\epsilon} \\ &\quad \times \frac{i}{q_1^2 - m^2 + i\epsilon} \frac{i}{(k_1 - q_1)^2 - m^2 + i\epsilon} \cancel{\frac{i}{k_2^2 - m^2 + i\epsilon}} e^{ik_1 x_1} e^{-ik_2 x_2} (2\pi)^4 \delta^4(k_1 - k_2) \\ &= (2\pi)^4 \delta^4(p_1 - p_2) \frac{(ig)^2}{2} \int \frac{d^4 q}{(2\pi)^4} \frac{i}{q^2 - m^2 + i\epsilon} \frac{i}{(p_1 - q)^2 - m^2 + i\epsilon} \\ &= (2\pi)^4 \delta^4(p_1 - p_2) i \mathcal{M} \end{aligned}$$

Momentum space Feynman rules

Therefore finally we can get

$$i\mathcal{M} = \frac{(ig)^2}{2} \int \frac{d^4q}{(2\pi)^4} \frac{i}{q^2 - m^2 + i\epsilon} \frac{i}{(p_1 - q)^2 - m^2 + i\epsilon}$$

We can summarize this procedure with the **momentum-space Feynman rules**. These Feynman rules, given in Box 7.1, tell us how to directly calculate $i\mathcal{M}$ from pictures. With these rules, you can forget about practically anything else we have covered so far.



Momentum space Feynman rules

Momentum-space Feynman rules

- Internal lines (those not connected to external points) get propagators $\frac{i}{p^2 - m^2 + i\epsilon}$.
- Vertices come from interactions in the Lagrangian. They get factors of the coupling constant times i .
- Lines connected to external points do not get propagators (their propagators are canceled by terms from the LSZ reduction formula).
- Momentum is conserved at each vertex.
- Integrate over undetermined 4-momenta.
- Sum over all possible diagrams.

Sign of momentum

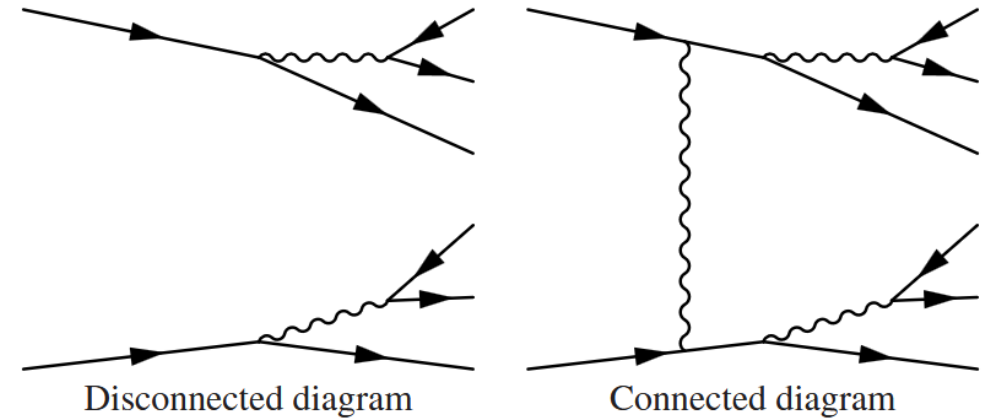
There is unfortunately no standard convention about how to choose the direction in which the momenta are going.

Momentum conservation : $\sum p_i = \sum p_f$

For internal lines, it does not matter if we use k^μ or $-k^\mu$. But you should keep momentum conservation at vertex.

Disconnected diagram

In disconnected diagrams,



$$\mathcal{M}_{\text{disc}} = \mathcal{M}_A \mathcal{M}_B$$

So our probability is

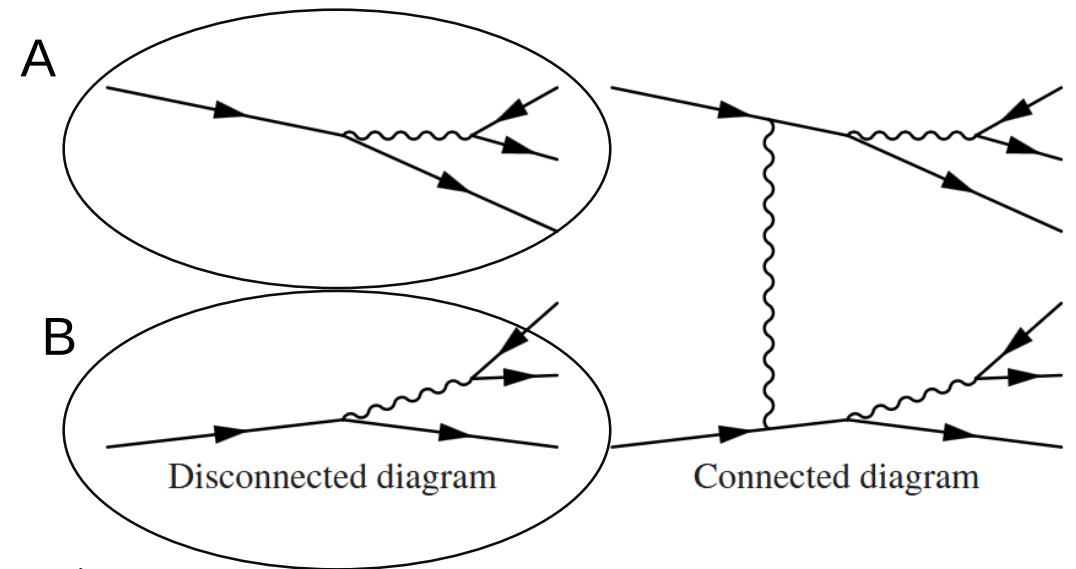
$$P_{\text{disc}} = P_A P_B$$

If we know one 1 to 3 diagram, then don't have to compute to compute 2 to 6 diagram which is composed by them. So, $P = P_{\text{conn}} + P_{\text{disc}} = P_{\text{conn}} + P_A P_B$

If disconnected and connected diagrams have same initial and final state, then Do they have interference?

Disconnected diagram

Nope. They don't have interference.



Connected diagram has one delta function; $\delta^4(p_i - p_f)$ While disconnected diagram has extra at least one delta function to make momentum conservation for each process ; $\delta^4(p_i - p_f)\delta^4(Q_i - Q_f)$. And in our notation, total momentum conservation is canceled. probability is

$$\begin{aligned}
 P &\sim |\mathcal{M}_{\text{connect}} + \delta^4(Q)\mathcal{M}_{\text{disconnect}}|^2 \\
 &\sim |\mathcal{M}_{\text{con}}|^2 + \delta^4(Q)^2|\mathcal{M}_{\text{disc}}|^2 + 2\text{Re}[\delta^4(Q)\mathcal{M}_{\text{con}}^*\mathcal{M}_{\text{disc}}] \\
 \frac{\delta^4(Q)}{\delta^4(Q)^2} &\sim \frac{1}{TV} \rightarrow 0
 \end{aligned}$$

The interference is suppressed in the asymptotic limit because the disconnected contribution contains an additional momentum-conserving delta function.

Example

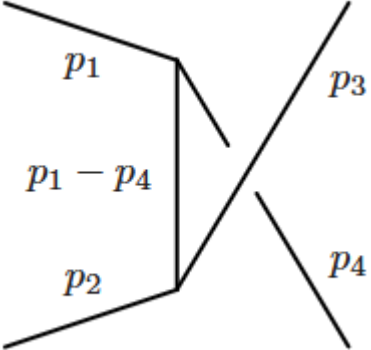
$$\mathcal{L} = -\frac{1}{2} \phi(\square + m^2)\phi + \frac{g}{3!} \phi^3$$

Consider $\phi\phi \rightarrow \phi\phi$.

$$i\mathcal{M}_s = \begin{array}{c} \text{Diagram: s-channel exchange} \\ \text{Two incoming lines from the left labeled } p_1 \text{ (top) and } p_2 \text{ (bottom).} \\ \text{They meet at a vertex, and a horizontal line labeled } p_1 + p_2 \text{ connects to a second vertex.} \\ \text{From this second vertex, two outgoing lines go to the right labeled } p_3 \text{ (top) and } p_4 \text{ (bottom).} \end{array} = (ig)^2 \frac{i}{(p_1 + p_2)^2 - m^2 + i\epsilon} = \frac{-ig^2}{s - m^2 + i\epsilon}$$

$$i\mathcal{M}_t = \begin{array}{c} \text{Diagram: t-channel exchange} \\ \text{Two incoming lines from the left labeled } p_1 \text{ (top) and } p_2 \text{ (bottom).} \\ \text{They meet at a vertex, and a vertical line labeled } p_1 - p_3 \text{ connects to a second vertex.} \\ \text{From this second vertex, two outgoing lines go to the right labeled } p_3 \text{ (top) and } p_4 \text{ (bottom).} \end{array} = (ig)^2 \frac{i}{(p_1 - p_3)^2 - m^2 + i\epsilon} = \frac{-ig^2}{t - m^2 + i\epsilon}$$

Example


$$i\mathcal{M}_u = \text{diagram} = (ig)^2 \frac{i}{(p_1 - p_4)^2 - m^2 + i\epsilon} = \frac{-ig^2}{u - m^2 + i\epsilon}$$

So,

$$i\mathcal{M} = i\mathcal{M}_s + i\mathcal{M}_t + i\mathcal{M}_u$$

$$\frac{d\sigma}{d\Omega} = \frac{g^4}{64\pi^2 s} \left[\frac{i}{s - m^2} + \frac{i}{t - m^2} + \frac{i}{u - m^2} \right]^2$$

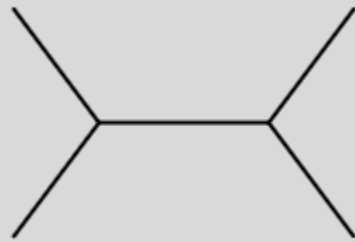
Mandelstam variables

$$s = (p_1 + p_2)^2 = (p_3 + p_4)^2$$

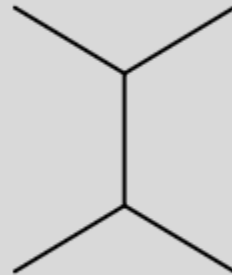
$$t = (p_1 - p_3)^2 = (p_2 - p_4)^2$$

$$u = (p_1 - p_4)^2 = (p_2 - p_3)^2$$

2 → 2 scattering channels.



s-channel



t-channel



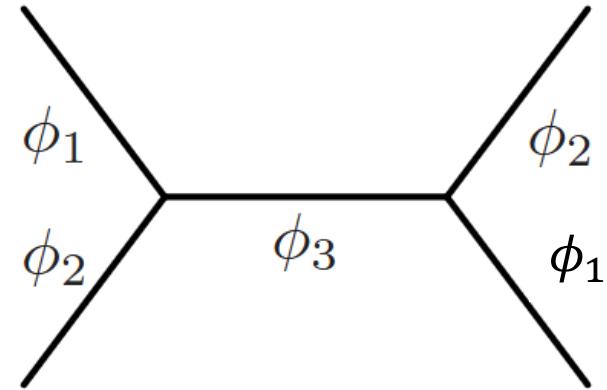
u-channel

Derivative couplings

For example,

$$\mathcal{L}_{\text{int}} = \lambda \phi_1 (\partial_\mu \phi_2) (\partial^\mu \phi_3)$$

with scalar ϕ_1, ϕ_2 and ϕ_3 .



$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{-ipx} + a_p^\dagger e^{ipx}). \quad (7.102)$$

So, if the particle is being created (emerging from a vertex) it gets a factor of ip_μ , and if it is being destroyed (entering a vertex) it gets a factor of $-ip_\mu$. So, we get a minus for incoming momentum and a plus for outgoing momentum. In this case, it is quite important to keep track of whether momentum is flowing into or out of the vertex.

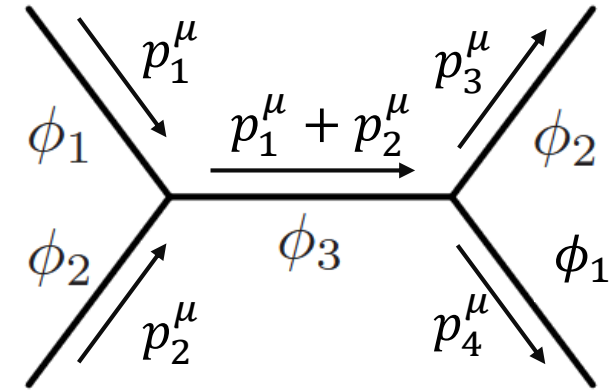


Derivative couplings

For example,

with scalar ϕ_1, ϕ_2 and ϕ_3 .

$$\mathcal{L}_{\text{int}} = \lambda \phi_1 (\partial_\mu \phi_2) (\partial^\mu \phi_3)$$



$$\langle \phi_2; p_3, \phi_1; p_4 | \phi_1 (\partial_\nu \phi_2) (\partial^\nu \phi_3) \phi_1 (\partial_\mu \phi_2) (\partial^\mu \phi_3) | \phi_1; p_1, \phi_2; p_2 \rangle$$

Diagram illustrating the matrix element calculation with momentum flow and creation/annihilation operators:

- Blue arrows indicate the flow of momenta p_3 and p_4 from the final state to the interaction vertex, labeled $a_{p_3} e^{-ip_3 x}$ and a_{p_4} .
- Red arrows indicate the flow of momenta p_1 and p_2 from the initial state to the interaction vertex, labeled a_{p_1} and $a_{p_2} e^{-ip_2 x}$.

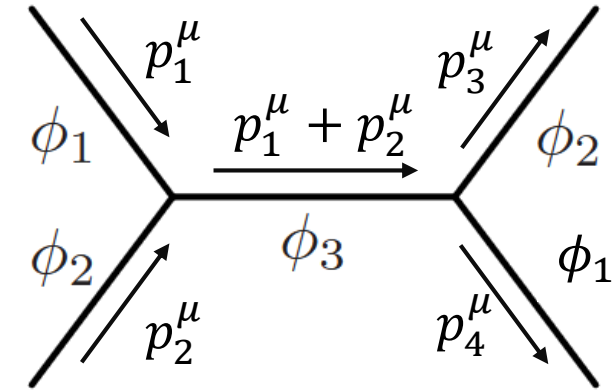


Derivative couplings

For example,

$$\mathcal{L}_{\text{int}} = \lambda \phi_1 (\partial_\mu \phi_2) (\partial^\mu \phi_3)$$

with massless scalar ϕ_1, ϕ_2 and ϕ_3 .



$$\langle \phi_2; p_3, \phi_1; p_4 | \phi_1 (ip_{3\nu} \phi_2) (\partial^\nu \phi_3) \phi_1 (-ip_{2\mu} \phi_2) (\partial^\mu \phi_3) | \phi_1; p_1, \phi_2; p_2 \rangle$$

$a_k e^{-i(p_1+p_2)x}$
 $a_k^\dagger e^{i(p_1+p_2)x}$

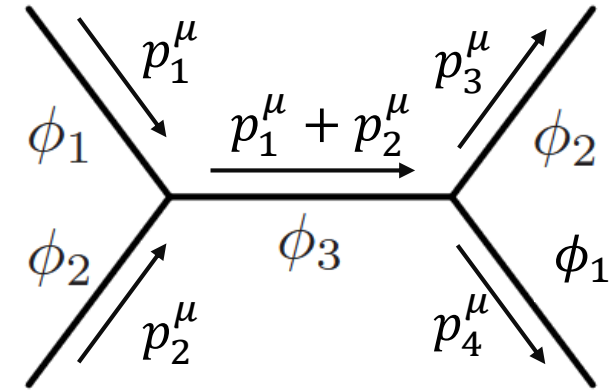


Derivative couplings

For example,

$$\mathcal{L}_{\text{int}} = \lambda \phi_1 (\partial_\mu \phi_2) (\partial^\mu \phi_3)$$

with scalar ϕ_1, ϕ_2 and ϕ_3 .



$$\langle \phi_2; p_3, \phi_1; p_4 | \phi_1 (i p_{3\nu} \phi_2) (-i(p_1^\nu + p_2^\nu) \phi_3) \phi_1 (-i p_{2\mu} \phi_2) (i(p_1^\mu + p_2^\mu) \phi_3) | \phi_1; p_1, \phi_2; p_2 \rangle$$

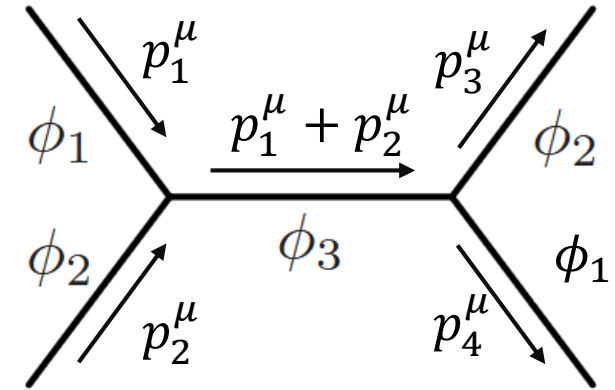
So

$$\begin{aligned} i\mathcal{M} &= (i\lambda)^2 (-i p_{2\mu}) \left(i(p_1^\mu + p_2^\mu) \right) \frac{i}{s} (-i(p_1^\nu + p_2^\nu)) (i p_{3\nu}) \\ &= -i\lambda^2 \frac{[p_1 p_2][p_3 p_4]}{s} \end{aligned}$$



Derivative couplings

As a cross check, we should get the same answer if we use a different Lagrangian related to the one we used by integration by parts



$$\mathcal{L}_{\text{int}} = \lambda \phi_1 (\partial_\mu \phi_2) (\partial^\mu \phi_3) = -\lambda \phi_3 [(\partial^\mu \phi_1) (\partial_\mu \phi_2) + \phi_1 \square \phi_2]$$

$$\begin{aligned} i\mathcal{M} &= (-i\lambda)^2 \left[(-ip_1^\mu) (-ip_{2\mu}) + (-ip_{2\mu})^2 \right] \frac{i}{s} [(ip_4^\nu) (ip_{3\nu}) + (ip_{3\nu})^2] \\ &= -i\lambda^2 \frac{[p_1 p_2] [p_3 p_4]}{s} \end{aligned}$$

Summary

$$\begin{aligned} \sigma? &\rightarrow \mathcal{M}? \rightarrow \langle f|S|i\rangle? \rightarrow \text{LSZ} \langle \Omega|T\{\phi \dots\}|\Omega\rangle? \rightarrow \text{LSZ} \langle 0|T\{\phi_0 \dots \exp(i \int d^4x \mathcal{L}_{\text{int}})\}|0\rangle_{\text{no bubble}}? \\ &\rightarrow \text{LSZ} (D_F D_F D_F D_F \dots) \rightarrow (\text{internal } D_F\text{'s}) \rightarrow \text{Momentum space} \dots \end{aligned}$$

Summary

We can summarize this procedure with the **momentum-space Feynman rules**. These Feynman rules, given in Box 7.1, tell us how to directly calculate $i\mathcal{M}$ from pictures. With these rules, you can forget about practically anything else we have covered so far.

$$\sigma? \rightarrow \mathcal{M}? \rightarrow \langle f|S|i\rangle? \rightarrow \text{LSZ} \langle \Omega | T \{ \phi \dots \} | \Omega \rangle \rightarrow \text{LSZ} \langle 0 | T \{ \psi_0 \dots \psi_0 \} | 0 \rangle \rightarrow \text{LSZ} (D_r \dots D_F \dots) \rightarrow (\text{internal } D_F \text{'s}) \rightarrow \text{Momentum space ...}$$

no bubble?

Momentum-space Feynman rules

- Internal lines (those not connected to external points) get propagators $\frac{i}{p^2 - m^2 + i\epsilon}$.
- Vertices come from interactions in the Lagrangian. They get factors of the coupling constant times i .
- Lines connected to external points do not get propagators (their propagators are canceled by terms from the LSZ reduction formula).
- Momentum is conserved at each vertex.
- Integrate over undetermined 4-momenta.
- Sum over all possible diagrams.

Summary

We can summarize this procedure with the **momentum-space Feynman rules**. These Feynman rules, given in Box 7.1, tell us how to directly calculate $i\mathcal{M}$ from pictures. With these rules, you can forget about practically anything else we have covered so far.

$$\sigma? \rightarrow \mathcal{M}? \rightarrow \langle f|S|i\rangle? \rightarrow \text{LSZ} \langle \Omega|T\{\phi \dots\}|\Omega\rangle \rightarrow \text{LSZ} \langle 0|T\{\psi_0 \dots \psi_0^{\dagger} \mathcal{L}_{\text{int}}\}|0\rangle \text{ 'no bubble' }?$$

$$\rightarrow \text{LSZ} (D_r \dots D_F \dots) \rightarrow (\text{internal } D_F \text{'s}) \rightarrow \text{Momentum space ...}$$

Momentum-space Feynman rules

- Internal lines (those not connected to external points) get propagators $\frac{i}{p^2 - m^2 + i\epsilon}$.
- Vertices come from interactions in the Lagrangian. They get factors of the coupling constant times i .
- Lines connected to external points do not get propagators (their propagators are canceled by terms from the LSZ reduction formula).
- Momentum is conserved at each vertex.
- Integrate over undetermined 4-momenta.
- Sum over all possible diagrams.

+ Mandelstam variable
+ Derivative coupling

Appendix

Appendix – LSZ reduction formula : spin 0

$$\begin{aligned} i \int d^4x e^{ipx} (\square + m^2) \phi(x) &= i \int d^4x e^{ipx} [\partial_t^2 - \vec{\nabla}^2 + m^2] \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} (a_k(t) e^{-ikx} + a_k^\dagger(t) e^{ikx}) \\ &= i \int d^4x e^{ipx} \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} [(\ddot{a}_k - 2i\omega_k \dot{a}_k - (\omega_k^2 + \vec{k}^2 + m^2) a_k) e^{-ikx} + h.c.] \\ &= i \int d^4x e^{ipx} \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} [(\ddot{a}_k - 2i\omega_k \dot{a}_k) e^{-ikx} + h.c.] \\ &= \frac{i}{\sqrt{2\omega_p}} \int dt [\ddot{a}_p - 2i\omega_p \dot{a}_p + e^{2i\omega_p t} (\ddot{a}_{-p}^\dagger + 2i\omega_p \dot{a}_{-p}^\dagger)] \\ &= \frac{i}{\sqrt{2\omega_p}} [\dot{a}_p - 2i\omega_p a_p + e^{2i\omega_p t} \dot{a}_{-p}^\dagger]_{-\infty}^{\infty} = \sqrt{2\omega_p} [a_p(\infty) - a_p(-\infty)] \end{aligned}$$

Appendix – LSZ reduction formula : spin $\frac{1}{2}$

$$\begin{aligned} & \int d^4x \bar{u}_s(p) e^{ipx} (i\partial - m) \psi(x) \\ &= \int d^4x \bar{u}_s(p) e^{ipx} (i\partial - m) \sum_r \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} (a_{k,r} u_r(k) e^{-ikx} + b_{k,r}^\dagger v_r(k) e^{ikx}) \\ &= \sum_r \int d^4x \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \bar{u}_s(p) (i\gamma^0 \dot{a}_{k,r} u_r(k) e^{-ikx} + i\gamma^0 \dot{b}_{k,r} v_r(k) e^{ikx}) \\ &= \frac{i}{\sqrt{2\omega_p}} \int dt \left[\sum_r \dot{a}_{p,r} u_s^\dagger(p) u_r(p) + e^{2i\omega_p t} \sum_r \dot{b}_{-p,r} u_s^\dagger(p) v_r(-p) \right] \\ &= i \sqrt{2\omega_p} (a_p(\infty) - a_p(-\infty)) \end{aligned}$$

Appendix – LSZ reduction formula : spin $\frac{1}{2}$

$$\int d^4x \bar{u}_s(p) e^{ipx} (i\partial - m) \psi(x) = i \sqrt{2\omega_p} [a_{p,s}(\infty) - a_{p,s}(-\infty)]$$

Similarly, we can calculate the others.

$$\int d^4x \bar{v}_s(p) e^{-ipx} (i\partial - m) \psi(x) = i \sqrt{2\omega_p} [b_{p,s}^\dagger(\infty) - b_{p,s}^\dagger(-\infty)]$$

Appendix – Lagrangian derivation

Our Lagrangian is

$$\mathcal{L} = -\frac{1}{2}\phi(\square + m^2)\phi + \mathcal{L}_{\text{int}}[\phi]$$

Interacting field satisfies Euler-Lagrange Eq,

$$(\square + m^2)\phi(x) = \mathcal{L}'_{\text{int}}[\phi]$$

and at equal time commutator,

$$\begin{aligned} [\phi(\vec{x}), \phi(\vec{y})] &= 0 \\ [\phi(\vec{x}), \dot{\phi}(\vec{y})] &= i\delta^3(\vec{x} - \vec{y}) \end{aligned}$$

Appendix – Lagrangian derivation

Consider $(\square_x + m^2)\langle\Omega|T\{\phi(x)\phi(y)\}|\Omega\rangle$. Time derivative make delta function because of theta function of time ordering operator.

$$\begin{aligned}(\square_x + m^2)\langle\phi(x)\phi(y)\rangle &= \langle(\square_x + m^2)\phi(x)\phi(y)\rangle - i\delta^4(x - y) \\ &= \langle\mathcal{L}'_{\text{int}}[\phi(x)]\phi(y)\rangle - i\delta^4(x - y)\end{aligned}$$

Generally,

$$\begin{aligned}(\square_x + m^2)\langle\phi(x)\phi(x_1)\cdots\phi(x_n)\rangle \\ = \langle\mathcal{L}'_{\text{int}}[\phi(x)]\phi(x_1)\cdots\phi(x_n)\rangle - i\sum_j\delta^4(x - x_j)\langle\phi(x_1)\cdots\phi(x_{j-1})\phi(x_{j+1})\cdots\phi(x_n)\rangle\end{aligned}$$

This is Schwinger – Dyson equation.

Appendix – Lagrangian derivation

As an example, let's consider free theory in $\langle 0|T\{\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\}|0\rangle$

$$\begin{aligned}\langle 0|\{\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\}|0\rangle &= \int d^4x \delta_{x_1} \langle 0|T\{\phi(x)\phi(x_2)\phi(x_3)\phi(x_4)\}|0\rangle \\ &= i \int d^4x [(\square_x + m^2)D_{x_1}] \langle 0|T\{\phi(x)\phi(x_2)\phi(x_3)\phi(x_4)\}|0\rangle \\ &= i \int d^4x D_{x_1}(\square_x + m^2) \langle 0|T\{\phi(x)\phi(x_2)\phi(x_3)\phi(x_4)\}|0\rangle \\ &= i \int d^4x D_{x_1} \delta_{x_2} \langle 0|T\{\phi(x_3)\phi(x_4)\}|0\rangle + \delta_{x_3} \langle 0|T\{\phi(x_2)\phi(x_4)\}|0\rangle \\ &\quad + \delta_{x_4} \langle 0|T\{\phi(x_2)\phi(x_3)\}|0\rangle \\ &= i \int d^4x (D_{12}D_{34} + D_{13}D_{24} + D_{14}D_{23})\end{aligned}$$

Appendix – Lagrangian derivation

In ϕ^3 interaction theory, $\mathcal{L}'_{\text{int}} = \frac{g}{2} \phi^2$

$$\begin{aligned}\langle \Omega | T \phi(x_1) \phi(x_2) | \Omega \rangle &= \langle \phi_1 \phi_2 \rangle \\ &= \int d^4x \delta_{x1} \langle \phi_x \phi_2 \rangle \\ &= i \int d^4x D_{x1} (\langle \mathcal{L}'_{\text{int}}[\phi_x] \phi_2 \rangle - i \delta_{x2}) \\ &= D_{12} + i \frac{g}{2} \int d^4x D_{x1} \langle \phi_x^2 \phi_2 \rangle \\ &= D_{12} - \frac{g}{2} \int d^4x d^4y D_{x1} \langle \phi_x^2 \mathcal{L}'_{\text{int}}[\phi_y] \rangle + ig \int d^4x D_{1x} D_{2x} \langle \phi_x \rangle \\ &= D_{12} - \frac{g^2}{4} \int d^4x d^4y D_{x1} \langle \phi_x^2 \phi_y^2 \rangle + ig \int d^4x D_{1x} D_{2x} \langle \phi_x \rangle\end{aligned}$$

Appendix – Lagrangian derivation

$$\begin{aligned}\langle \Omega | T \phi(x_1) \phi(x_2) | \Omega \rangle &= D_{12} - \frac{g^2}{4} \int d^4x d^4y D_{x1} \langle \phi_x^2 \phi_y^2 \rangle + ig \int d^4x D_{1x} D_{2x} \langle \phi_x \rangle \\ &= D_{12} - \frac{g^2}{4} \int d^4x d^4y D_{x1} \left(2D_{xy}^2 + D_{xx} D_{yy} + \mathcal{O}(g) \right) \\ &\quad - g \int d^4x D_{1x} D_{2x} \left(\frac{g}{2} \int d^4y D_{xy} D_{yy} + \mathcal{O}(g) \right) \\ &= D_{12} - g^2 \int d^4x d^4y \left(\frac{1}{2} D_{1x} D_{xy}^2 D_{y2} + \frac{1}{4} D_{1x} D_{xx} D_{yy} D_{y2} + \frac{1}{2} D_{1x} D_{2x} D_{xy} D_{yy} \right)\end{aligned}$$

The position-space Feynman rules that connect this expansion to diagrams are the same as those coming from the Hamiltonian approach

Appendix – Wick Theorem

Split the field as creation and annihilation parts,

$$\phi = \phi^- + \phi^+ = \int \frac{d^3p}{(2\pi)^3} a_p e^{-ipx} + \int \frac{d^3p}{(2\pi)^3} a_p^\dagger e^{ipx}$$

And define normal ordering : Creation on left, Annihilation on right. For example, Normal ordered two fields is

$$:\phi(x)\phi(y): = \phi^+(x)\phi^+(y) + \phi^+(x)\phi^-(y) + \phi^+(y)\phi^-(x) + \phi^-(x)\phi^-(y)$$

Normal ordered things always vanish in the vacuum except they are c -number functions.

$$\langle 0|: \phi \dots \phi :|0\rangle = 0, \quad \langle 0|: c :|0\rangle = c$$

Appendix – Wick Theorem

Let's consider two point function ; $T\{\phi(x_1)\phi(x_2)\}$

$t_1 > t_2$ case,

$$\begin{aligned} T\{\phi(x_1)\phi(x_2)\} &= \phi^+(x_1)\phi^+(x_2) + \phi^+(x_1)\phi^-(x_2) + \phi^-(x_1)\phi^+(x_2) + \phi^-(x_1)\phi^-(x_2) \\ &= \phi^+(x_1)\phi^+(x_2) + \phi^+(x_1)\phi^-(x_2) + \phi^+(x_2)\phi^-(x_1) + [\phi^-(x_1), \phi^+(x_2)] + \phi^-(x_1)\phi^-(x_2) \\ &=: \phi(x_1)\phi(x_2): + [\phi^-(x_1), \phi^+(x_2)] \end{aligned}$$

therefore,

$$\begin{aligned} T\{\phi(x_1)\phi(x_2)\} &= \phi^+(x_1)\phi^+(x_2) + \phi^+(x_1)\phi^-(x_2) + \phi^-(x_1)\phi^+(x_2) + \phi^-(x_1)\phi^-(x_2) \\ &= \phi^+(x_1)\phi^+(x_2) + \phi^+(x_1)\phi^-(x_2) + \phi^+(x_2)\phi^-(x_1) + [\phi^-(x_1), \phi^+(x_2)] + \phi^-(x_1)\phi^-(x_2) \\ &=: \phi(x_1)\phi(x_2): + \theta(t_1 - t_2)[\phi^-(x_1), \phi^+(x_2)] + \theta(t_2 - t_1)[\phi^-(x_2), \phi^+(x_1)] \end{aligned}$$

Appendix – Wick Theorem

And we know, the second and third terms give us Feynman propagator in free vacuum. For convenience, we denote

$$T\{\phi(x_1)\phi(x_2)\} =: \phi(x_1)\phi(x_2): + D_F(x_1, x_2)$$

The Wick theorem guarantees

$$T\{\phi(x_1) \cdots \phi(x_n)\} =: \phi(x_1) \cdots \phi(x_n): + \text{all possible contraction}$$

I could prove it, and I still have plenty of paper left, but I'm too tired for that, so I'll call it a day here